

QUESTIONS:

1. Explain the potential merits of using a Round Robin algorithm vs. first-come-first-serve scheduling algorithm. (4 points)

2. A system adopts the '**Round Robin**' scheduling with a **Time Quantum=4** wherein the Arrival Time and Burst Time for each process has been shown in the table below. In a recorded time span, the system has seven processes, P_1 , P_2 , P_3 , P_4 , P_5 , P_6 and P_7 as shown in the following table. All the I/O operations are carried out on different devices.

At time $t=14$, the process running will ask for some I/O. The I/O operation takes 14 time units to complete.

At time $t=16$, the process running will ask for some I/O. The I/O operation takes 14 time units to complete.

At time $t=38$, the process running will ask for some I/O. The I/O operation takes 7 time units to complete.

At time $t=61$, the process running will ask for some I/O. The I/O operation takes 7 time units to complete.

KINDLY NOTE:

- In case you encounter a situation wherein at time $t = x$, process P_A finishes its time quantum and an I/O operation is requested at the same time, then under such circumstances, P_A should be sent to the blocked queue (and not the next process in the ready queue).

Process ID	Arrival Time	Burst Time
P_1	$t=0$	15
P_2	$t=7$	13
P_3	$t=9$	21
P_4	$t=41$	9
P_5	$t=31$	8
P_6	$t=35$	7
P_7	$t=51$	14

A) Draw a Gantt Chart (Timing Diagram) for the above data.

(4 points)

B) Calculate the Waiting Time for each process.

(4 points)

20

C) Calculate the Response Time for each process.

(4 points)

D) Calculate the Turn-Around Time for each process.

(4 points)